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Downstream impacts of mines on agriculture in Africa[☆,☆☆]

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ABSTRACT

Mining operations in Africa are expanding rapidly, creating negative externalities that remain poorly understood. In this paper, we provide causal evidence for the impact of water pollution from mines on downstream vegetation and agriculture across the continent. We exploit a natural experiment, where mines cause a discontinuity in water pollution along river networks, to compare vegetation health in upstream and downstream locations. We find that mines significantly reduce peak vegetation downstream by 1.3–1.5%, impairing the productivity of over 74,000 km^2 of croplands. These reductions correspond to annual losses of 91,000–205,000 tons of cereal crops in the immediate vicinity alone, with particularly severe effects in fertile regions and areas where gold mining predominates. Our findings highlight substantial externalities of mining and demonstrate an urgent need for oversight and regulation.

1. Introduction

The mining sector in Africa is experiencing an unprecedented boom, driven by the demands of economic growth and the clean energy transition (Pörtner et al., 2022). This brings economic opportunities but also creates substantial negative externalities. Mining has been linked to pollution, environmental degradation, and biodiversity loss, as well as corruption, conflict, and child labor that undermine the livelihoods of local communities.[2] These externalities are striking and may have far-reaching impacts that could offset the economic benefits of mining, particularly in regions where extraction occurs. Insights into these impacts, however, remain scarce — especially in developing countries where weak institutions struggle to manage the impacts of mining, and external costs often go unmitigated.

In this paper, we provide causal evidence for the impacts of water pollution from mines on vegetation and agriculture in Africa. We exploit a natural experiment for identification, using the discontinuous jump in water pollution at mine locations along directed river networks. This leads to a sharp drop in vegetation health downstream of mines, while upstream areas remain unaffected and can serve as control groups. To implement this regression discontinuity design, we analyze outcomes across fine-grained river basins. We measure vegetation health and agricultural productivity across Africa by using satellite-derived peak vegetation indices combined with land use masks. This approach allows us to estimate the causal effects of water pollution from mines at an unprecedented geographic scale while maintaining rigorous causal identification.

Our research design identifies impacts of mines on vegetation that are transmitted through water.[3] The primary mechanism is water pollution: mines cause acidification, elevated salinity, and heavy metal contamination. These affect vegetation and agricultural productivity by disrupting plant physiology, soil microbiomes, and nutrient uptake. Secondary mechanisms involve farmers' adaptation in response to water pollution, such as relocating or switching crops.

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[2] See, for instance, Aska et al. (2024), Berman et al. (2017, 2023), Giljum et al. (2022), Girard et al. (2025), Knutsen et al. (2017), Macklin et al. (2023) and Santana et al. (2020).

[3] Our identification strategy has notable parallels in the literature. Most closely related, Gittard and Hu (2024) investigate the impacts of water pollution from industrial mining on health outcomes in Africa. Sigman (2002) and Lipscomb and Mobarak (2017) document water pollution spillovers across and within national borders, while Dias et al. (2023) and Strobl and Strobl (2011) use similar river-based discontinuities to study agricultural impacts in Brazil and Africa.

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Importantly, our research design controls for other impacts of mining — such as air pollution or economic effects — that do not follow river flows, allowing us to directly isolate the water channel.

The main contribution of our paper is the causal estimate of water-mediated mining impacts on vegetation and agriculture — an important externality that has previously been neglected. Earlier research has investigated various other impacts of mining[4] and many studies document pollution from mine sites.[5] While hydrological studies suggest that extensive areas and millions of livelihoods are affected by mining-induced river pollution (Macklin et al., 2023), causal evidence on this channel remains scarce and limited in scope. This paper addresses this gap by providing causal evidence for negative effects on vegetation across diverse mining contexts throughout Africa.

Our study contributes to several related literatures. Research on water pollution in developing contexts has typically focused on drinking water (e.g., Olmstead, 2010). We extend this literature by highlighting that clean water functions as a broad ecosystem service, supporting vegetation health and agricultural productivity (Keeler et al., 2012; Van Vliet et al., 2017). Our work also contributes to the growing body of research using remote-sensing methods to assess mining impacts in a data-scarce environment (Giljum et al., 2022; Maus and Werner, 2024), demonstrating how satellite data can enable causal analysis where traditional data sources are unavailable. Additionally, our findings relate to global environmental justice, as mineral extraction in lower-income African countries often serves demand from higher-income nations while imposing local environmental costs (see, e.g., Banzhaf et al., 2019).

We find that mines have substantial negative effects on downstream vegetation. Overall, peak vegetation indices decline by approximately 1.3 percent, while croplands experience reductions of about 1.5 percent. These effects are economically meaningful given their geographic scope — even the immediately impacted area, within 33 km downstream of mines, stretches over 280,000 km^2, including 74,000 km^2 of croplands. We estimate that these reductions cause annual losses of 91,000–205,000 tons of cereal crops. Moreover, we provide suggestive evidence that effects may extend much further downstream. Analysis of impact heterogeneity reveals that the largest negative effects occur in fertile regions and areas dominated by gold mining.

Our findings have important practical implications. Agriculture plays a vital role in local livelihoods and economies across Africa, and results show that large, fertile, and populated areas are affected by water pollution from mining. These impacts can cause economic and nutritional disruptions, potentially exacerbating food insecurity challenges across the continent. From a policy perspective, our study highlights inadequate oversight of the mining industry and the scarcity of data to assess their impacts.[6] Remote-sensing approaches can help address data gaps efficiently but rely on conventional data for calibration and are constrained in the types of effects that can be detected.

The remainder of this paper proceeds as follows. In the next section, we provide background and intuition for our analysis. In Section 3, we describe our data and empirical strategy. Section 4 presents the main results, heterogeneity analysis, and robustness checks. We discuss our results and their implications in Section 5 and conclude in Section 6.

2. Background and intuition

This section introduces key components of our study and explains the intuition behind our research design. We examine how mining operations affect vegetation and agriculture through water pollution, describe our unit of observation, river basins, and the satellite-derived vegetation indices.

2.1. Mines and mining in Africa

Mineral extraction is a growing sector in many African economies, presenting both opportunities and challenges. While mining can generate employment and stimulate local economic development (Bazillier and Girard, 2020; Gräser, 2024; von der Goltz and Barnwal, 2019; Ofosu et al., 2020), it also brings substantial risks. Resource wealth may crowd out other industries, drive corruption and conflict (Berman et al., 2017, 2023; Knutsen et al., 2017), and mining drives rapid environmental changes (Aska et al., 2024; Barenblitt et al., 2021; Giljum et al., 2022; Girard et al., 2025) and pollution with considerable impacts on people and their environments (Awotwi et al., 2021; Macklin et al., 2023).

Vegetation and agriculture are affected through multiple channels, with water pollution playing a particularly important role (Santana et al., 2020). Mining requires copious amounts of water for processing and disturbs orders of magnitude more material than the metal eventually extracted. This creates erosion and generates waste materials, including sediment, rock, and heavy metals, that are stored as 'tailings'. These tailings, oxidized by air and weathered by rain, steadily cause pollution of water resources as they feed into rivers (Schwarzenbach et al., 2010) and are well-known ecological risks (Wu et al., 2023). Combined with the release of toxic chemicals used for processing,[7] mining can induce water pollution from heavy metals (Frossard et al., 2018), increase salinity (Russ et al., 2020; Zörb et al., 2019), and cause acidification (Du et al., 2024) — all of which impact vegetation health via plant physiology and growth.[8] These impacts can persist for hundreds of years, degrade water quality, and devastate terrestrial and aquatic ecosystems (Du et al., 2024; Barrie Johnson and Hallberg, 2005).

Mines also affect vegetation in ways other than water pollution. Air pollution, particularly from coal mining and processing of metals in smelters, is a prominent example (Fugiel et al., 2017; Miao et al., 2017; Pandey et al., 2014). For Ghana, Aragón and Rud (2015) find that air pollution from gold mining in Ghana reduced total factor productivity in the agricultural sector by 40%, mostly through direct effects on crop health by air and the resulting soil pollution. Socioeconomic factors constitute another pathway through which mines affect agriculture and vegetation. Corruption, conflict, and weak institutions have been linked to mining and are detrimental to agricultural productivity (Wuepper et al., 2023). Pollution also has negative impacts on labor supply (Hanna and Oliva, 2015) and productivity (Zivin and Neidell, 2012), human capital accumulation (Currie et al., 2009), and mining industries affect local income and employment (see, e.g., Bazillier and Girard, 2020; Gräser, 2024; Kotsadam and Tolonen, 2016). These impacts may, in turn, affect agricultural productivity.

The risks of mining are particularly acute in Africa, where oversight is often lacking (Macklin et al., 2023). International regulations for the mining industry have been scarce,[9] while the prevalent small-scale, artisanal mines often operate outside regulatory frameworks. Countries in Africa often lack the robust institutional frameworks necessary to monitor and regulate expanding mining operations and face severe challenges in managing detrimental externalities. Concerns are further emphasized by the heavy reliance on subsistence agriculture, looming food insecurity, and the potential long-term consequences of environmental degradation (see, e.g., Audry et al., 2004; Barrie Johnson and Hallberg, 2005).

[4] Aragón and Rud (2015), Aska et al. (2024), Berman et al. (2017), Gittard and Hu (2024), Ofosu et al. (2020), von der Goltz and Barnwal (2019)

[5] Awotwi et al. (2021), Du et al. (2024), Duncan (2020), Mulenga et al. (2024), Wu et al. (2023)

[6] Also see Auld et al. (2018), Maus and Werner (2024) and Jones et al. (2024).

[7] Sodium cyanide and mercury, for example, are heavily used in the (artisanal) extraction of gold (Duncan, 2020; Malone et al., 2023; Verbrugge et al., 2021) and can cause long-term alterations of sediment chemistry.

[8] See Appendix B1 for a literature overview on the effects of water pollution on plant health.

[9] A recent exception is the 2020 Global Industry Standard on Tailings Management (GISTM; UNEP, 2023), which was established to prevent future tailings dam failures after the catastrophic dam collapse at Vale's Corrego de Feijao mine in Brazil.

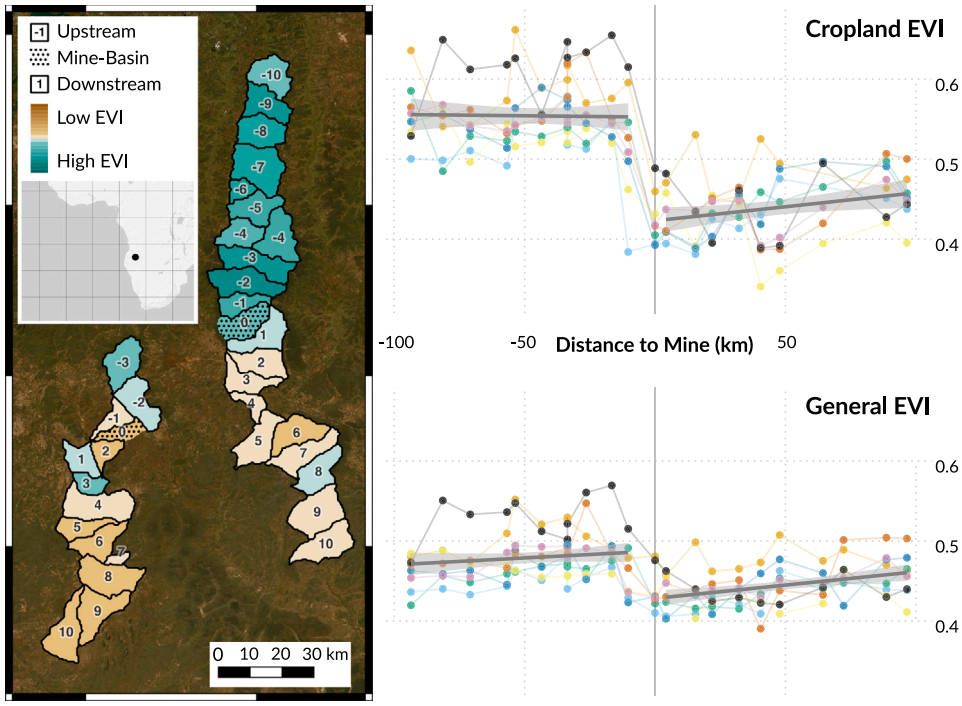

Fig. 1. Example of two Angolan mine sites (dotted, and labeled with '0') and their upstream and downstream basin systems (left), as well as measurements of the Enhanced Vegetation Index (EVI) for croplands and general vegetation over the years 2016, 2017, 2018, 2019, 2020, 2021, 2022, and 2023 along the larger basin system (right).

2.2. River basins and water streams

Our unit of observation is the river basin, defined as the area drained by a river and its tributaries. Basin boundaries are defined by topographic features, like ridges and hills, ensuring that all surface waters converge to a single point.[10] This definition can be applied recursively, and basins at local to continental scales can be nested in a hierarchical system. Our analysis uses the finest scale of basins (in the dataset of Lehner and Grill, 2013), illustrated in the left panel of Fig. 1. Their small size conveys localized impacts while ensuring that relevant factors vary smoothly and discontinuities can be attributed to mine sites themselves. Coupled with the unidirectional flow of water down river networks, this allows us to implement our design: downstream basins may be affected by contaminated water, but upstream basins remain unaffected and can serve as controls.

2.3. Agricultural productivity from space

We use satellite-derived vegetation indices as proxies for vegetation health and agricultural productivity. In our context, where smallholder farming predominates, traditional datasets based on surveys or censuses face considerable challenges and lack consistent coverage across African countries. We construct peak vegetation indices, which correlate strongly with gross primary production of vegetation (Shi et al., 2017; Johnson, 2016) and are frequently used in similar analyses (see, e.g., Asher and Novosad, 2020; Wuepper et al., 2023; Boudot-Reddy and Butler, 2025), using vegetation and cropland masks. These consistently capture outcomes of interest with high granularity across time and space. An example of both masked measurements is provided in Fig. 1, demonstrating how the indices capture vegetation patterns up- and downstream of a mining site.

[10] Two prominent examples of large basins are the Mississippi River, which drains most of the mainland United States into the Gulf of Mexico, and the Congo Basin, which covers most of the Democratic Republic of the Congo (DRC) and drains into the Atlantic.

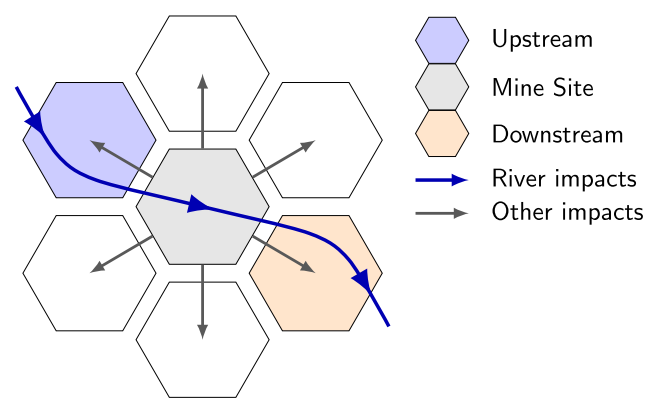

Fig. 2. Illustration of the research design. The comparison of up- and downstream basins enables the identification of mine impacts that are mediated by the river.

2.4. Research design

Our empirical strategy integrates mining sites, river basins, and remotely sensed vegetation indices to identify causal effects of mines on downstream agricultural productivity and vegetation health across the African continent. Fig. 2 illustrates our approach. We identify basins containing mine sites and compare vegetation in basins located upstream and downstream within river networks. This allows us to isolate the directed river-mediated impact by controlling for undirected effects of mines, thereby identifying the causal effect mediated by water flows.

Our strategy relies on the assumption that the observed discontinuities along rivers stem from mines themselves. This is plausible when mine locations are quasi-random rather than driven by factors related to vegetation or agriculture. The location of mines is mainly determined by accessible mineral deposits. Potential confounders exist (e.g., strategic placement near transportation) but are unlikely to confound our

estimates because their impacts would need to align with water flow direction and affect mine placement at the basin level.

We primarily focus on the immediate vicinity of mines, where basin characteristics are most comparable and our research design delivers strong causal identification. The reach of impacts over greater distances presents additional challenges, as fundamental differences may emerge from geographical patterns. For instance, other pollution sources are more likely located downstream where river transport is economically attractive. Such factors are unlikely to affect the discontinuity that we exploit for identification but could confound our assessment of impact decay.

The research design captures mining impacts on vegetation mediated through water systems, offering a focused but partial view of externalities. Direct impacts on-site and impacts that are not directed along the river (such as dust deposition, employment effects shifting labor, or income effects allowing for fertilizer use) are not captured by our design. Instead, mediators for the effect of interest include both direct and indirect effects of water pollution. While downstream vegetation is immediately affected, human responses to these impacts, resulting soil pollution, and other second-order effects are also reflected in our estimates.

3. Materials and methods

This section describes the dataset and methods used. First, we describe the basin-level dataset and how it is constructed, and elaborate on individual variables included. Then, we present our empirical strategy.

3.1. Dataset

Our dataset is a balanced panel of $N = 14{,}334$ river basins in Africa, observed over eight years from 2016–2023, delineated by Lehner and Grill (2013).[11] For our analysis, we classify basins into three categories:

1. mine-basins, with a mine site in their catchment area,
2. downstream basins, which are downstream of *any* mine-basin,
3. upstream basins, which are upstream of *all* connected mine-basins.

We consider up- and downstream chains of basins (departing from mine-basins) with a maximum order of ten which lie at an average river distance of 105 km.

We observe a total of 1900 mine-basins, 6307 upstream and 6127 downstream basins that are visualized in Fig. 3. (For a summary of orders and distance of basins, see Table E3 in the Appendix.) Their average size is 120 km^2, for a total area of 1,701,343 km^2. Each basin is unique but may be related to multiple mine-basins. In this case, we associate the basin with the nearest mine-basin. The next (i.e., order one) downstream basin is unique for each basin, but there may be multiple upstream basins of any given order. Not all mine-basins have a full set of up- or downstream basins, and the number decreases with order (see Figure D7 for an overview). More details on the basin data are provided in Appendix A.

Summary statistics. Table 1 presents summary statistics for the most important variables used.[12] This includes vegetation indices, geographical information on elevation and slope, meteorological information on temperature and precipitation, as well as population and accessibility to urban areas. Variables are mapped from a raster level to the level of basins, with the exact procedures described below.

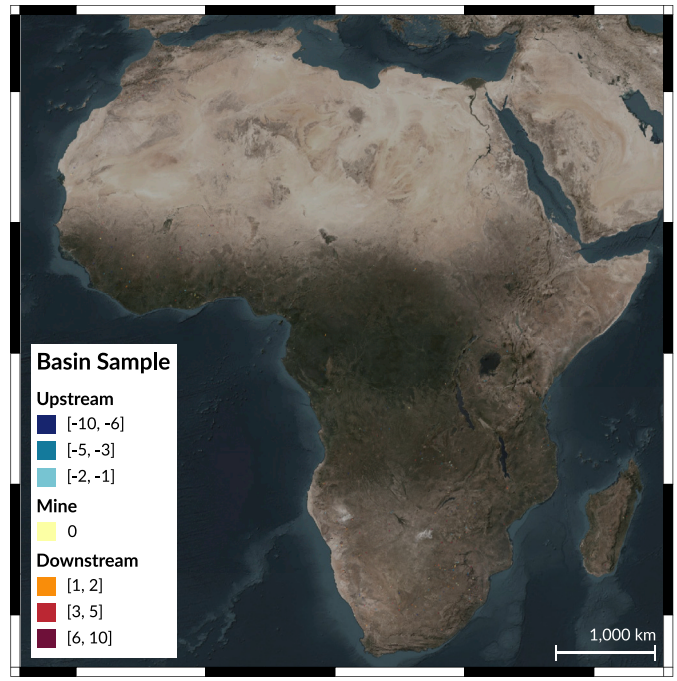

Fig. 3. Basins in the sample and their treatment status. Basemap imagery provided by Esri, Maxar, Earthstar Geographics, and the GIS User Community.

3.1.1. Vegetation and agriculture

To measure agricultural productivity and vegetation health, we create peak vegetation indices for areas with general and cropland-specific vegetation. We base our measures on the remotely sensed Enhanced Vegetation Index (EVI), which we combine with appropriate land use masks for vegetation and agriculture.[13]

We process the raw EVI data (Didan, 2015, which is available every 16 days at a 250 m resolution) by (1) applying quality filters to minimize the impact of clouds, (2) masking with yearly land use from the ESA CCI (Defourny et al., 2024, at a 300 m resolution)) to identify relevant areas for the two outcome measures, (3) aggregating the mean EVI at 16-day intervals per basin, and (4) computing the annual maximum EVI as a peak vegetation index. The resulting peak indices correlate strongly with measures of gross primary production of vegetation (see Shi et al., 2017, for an assessment) and crop yields (see, e.g., Azzari et al., 2017; Johnson, 2016). We consider several alternatives, including different land use masks (Digital Earth Africa, 2022) and aggregation schemes, as robustness checks.

3.1.2. Treatment

Our treatment is the incidence of a local mining site, derived from Maus et al. (2022). Their dataset integrates multiple sources, including the SNL Metals & Mining database, and provides comprehensive coverage of both industrial and artisanal mining sites. It offers two key advantages: (1) locations are verified through satellite imagery (2) mining areas (including features such as tailings dams, waste rock dumps, water ponds, and processing plants) are delineated within 10-km buffers around known locations. This way, smaller artisanal mining operations that often develop near larger industrial sites are

[11] The dataset has a hierarchical structure with twelve nested levels, where basins at each level are of comparable size. We use the finest Level 12, with 241,026 basins on the African continent that cover an average ($\{5, 50, 95\}$th percentile) area of 124.4 $(11.6, 131.4, 218.9)$ km^2, and feature an average diameter of 12.6 km and elevation differences of eleven meters.

[12] See Table E4 in the Appendix for summary statistics that are split by treatment status.

[13] The EVI's advantages (see Zeng et al., 2022, for a recent review) include high spatial granularity, frequent availability, and consistency over time and space. Moreover, it is less susceptible to atmospheric distortions and background noise than the commonly used Normalized Difference Vegetation Index (NDVI) and can accurately convey variation in low as well as high biomass conditions (Huete et al., 2002).

Table 1
Summary statistics for the basin-level dataset.

Variable	Unit	NT	Mean	St. Dev.	Min.	Max.
Peak vegetation	Index $[-1, 1]$	110,576	0.428	0.154	0.016	0.993
Mean vegetation	Index $[-1, 1]$	110,576	0.279	0.112	−0.021	0.578
Peak cropland Veg.	Index $[-1, 1]$	93,036	0.464	0.133	−0.068	0.978
Mean cropland Veg.	Index $[-1, 1]$	93,036	0.298	0.101	−0.104	0.601
Elevation	Meters	110,568	820.4	481.1	−118.3	3059.7
Slope	Degrees	110,568	2.23	2.34	0.0	20.9
Max. Temperature	Degree Celsius	110,572	34.3	3.9	15.6	48.8
Precipitation	Millimeters	110,576	901.8	595.2	0.64	4456.7
Population	Capita	110,576	8471	37,716	0.0	1,396,921
Accessibility	Minutes	110,528	164.3	179.1	1.0	2659.9

captured, and potential attenuation from missing or misclassified mine locations is reduced. The dataset lacks detailed mine characteristics and temporal information beyond the 2019 anchor, which we address via supplementary sources (inter alia, Jasansky et al., 2023; Padilla et al., 2021; Sepin et al., 2025; S&P Global Market Intelligence, 2025).

We use data on mining sites to construct our treatment variable as the distance to the nearest upstream mine-basin. For our primary specification, we consider the ordinal position of basins relative to the mine-basin as the treatment variable; as an alternative, we use different specifications based on river distance. We work with basins up to order ten but focus our analysis on the immediate vicinity (i.e., the first three basins in either direction) for comparability. The mine-basin itself can be considered treated, but on-site effects are not identified by our research design and thus not interpreted by us.

3.1.3. Covariates

We consider several covariates that could confound our analysis or provide additional insights. Our selection focuses on geographical characteristics that relate directly to river basins and variables with potential imbalances across treatment and control groups.

Geographic controls. For elevation and slope — key variables that define water flows and basin boundaries — we use high-resolution gridded data (Amatulli et al., 2018), aggregating from 30 arcsecond resolution (approximately 802–926 m) to basin-level averages. We determine the primary soil type for each basin using 250-m resolution data from the SoilGrids project (Hengl et al., 2017), to account for soil characteristics that influence vegetation health and may vary systematically along river networks.

Climate and socioeconomic controls. These factors become important at larger distances from mine sites, where treatment and control groups may diverge systematically. For the climate, we use precipitation data from CHIRPS (Dinku et al., 2018; Funk et al., 2015), and temperature data from TerraClimate (Abatzoglou et al., 2018), with meteorological data from CRU serving as an alternative for robustness checks (Harris et al., 2020). Both primary sources provide monthly measurements at 4–5 km resolution. We use annual precipitation totals and maximum monthly temperatures, following established practices in the literature. For socioeconomic conditions, we include total population (WorldPop, 2018) and average accessibility, measured by the travel time to the nearest city (Weiss et al., 2018), for each basin. Both are aggregated from a 1-km resolution using values from 2015 to control for initial conditions.

Other data. We also investigate relations to the number of violent events from ACLED (Raleigh et al., 2010), average annual air pollutant concentrations (Shen et al., 2024), nighttime lights (Elvidge et al., 2017), rates of forest loss (Hansen et al., 2013), and water pollution (United Nations Environment Programme, 2025).

3.1.4. Heterogeneity

We investigate heterogeneity of impacts across four main dimensions. First, we consider the Ecoregion classification (Dinerstein et al., 2017), which divides regional ecosystems with different types of agriculture and vegetation. We also consider country groups based on their primary crops and varying crop calendar cycles following the US Department of Agriculture (USDA).[14] Second, we proxy for mining intensity via the total mine area per mine-basin, and use the development of mining areas over time to proxy for activity.[15] Third, we investigate impacts by mineral type, collating various data sources to predict the most likely commodities mined in each basin (detailed in Appendix C2). Lastly, we examine heterogeneity across agricultural suitability based on the Global Agro-Ecological Zones (GAEZ) modeling framework (Fischer et al., 2021). For each basin, we calculate the average suitability for thirty crop types and categorize basins into high, medium, and low productivity based on the suitability of the best-suited crops.

3.2. Empirical strategy

We employ a regression discontinuity design (RDD) to identify the causal effects of mines on vegetation and agricultural productivity downstream. Our identification strategy exploits the natural discontinuity in the exposure to pollution along river networks. Basins downstream of mines are exposed to contaminated water, while upstream basins remain unaffected and can serve as controls.

Our application differs from standard RDDs in minor but notable ways. Our running variable — the distance to the mine-basin — has a natural direction and coincides with the treatment intensity. This directional feature strengthens our identification strategy, while the latter presents an opportunity to investigate impact decay. We also observe outcomes at the mine-basin, which is immediately impacted by contaminated water but also by third effects of the mine. These on-site observations provide useful information and suggestive evidence for other impacts of mine sites, but they are not identified by our research design and, hence, not interpreted.

Let x denote the *directed distance* from the mine along the river, where we have $x = 0$ at the mine-basin and $x < 0$ before it. Following the notation of Gelman and Imbens (2019), we can express the treatment effect as

$$\tau(x) = \mathrm{E}\left[y_i(1) - y_i(0) \mid x_i = x\right],$$

[14] The classification (available at ipad.fas.usda.gov) divides the continent into North Africa, Southern Africa, East Africa, and West Africa; due to limited observations, we merge North and East Africa.

[15] For the latter, we rely on a longitudinal extension of the mine dataset by Sepin et al. (2025), who automatically delineate mining sites over time using a segmentation model and high-resolution satellite imagery. The temporal information allows us to identify active mines based on their growth and assess their specific impacts.

where $y_i(1), y_i(0)$ are potential outcomes of observation i under treatment and control conditions. We exploit variation in x for identification and estimate the average treatment effect using the difference in conditional means between downstream and upstream locations:

$$\tau(x)_{ATE} = \mathrm{E}\left[y_i^{\text{obs}} \mid x_i = x, x > 0\right] - \mathrm{E}\left[y_i^{\text{obs}} \mid x_i = x, x < 0\right].$$

Our estimation equation is

$$y_{imt} = \boldsymbol{\beta}' F(x_i) + \boldsymbol{\theta}' W_{it} + \mu_m + \psi_t + \varepsilon_{imt}, \tag{1}$$

where y_{imt} represents vegetation health in basin i near mine m at time t, and x_i is the running distance. The vector W_{it} contains basin characteristics,[16] while μ_m and ψ_t represent mine- and year-fixed effects. The error term ε_{imt} is clustered by mine. We operationalize distance via the function $F(\cdot)$ that returns a vector separating upstream, downstream, and mine-basins.

3.2.1. Specification

Our preferred specification uses the basin order as distance, i.e., $x_i \in \{-10, \dots, 10\}$, and operationalizes it via indicators. This approach highlights basin-to-basin discrepancies and remains agnostic about functional forms, instead relying on local information at the level of our observations. Specifically, we let $F(\cdot)$ return indicators

$$f(x)_j = \mathrm{I}(x = j) \text{ for } j \in \{-10, \dots, -2, 0, 1, \dots 10\}.$$

Here, we omit the first upstream basin (order -1) as the reference category. Our design identifies differences between estimates at equivalent (absolute) distance (e.g., $\tau(8) = \beta_8 - \beta_{-8}$), which we report as effects of interest. For ease of interpretation, we also report pooled estimates comparing the first three basins.[17]

To assess the decay of impacts systematically, we specify an alternative model using the river distance in kilometers as the running variable. We operationalize this as

$$F(x) = \exp\{-\gamma \times |x|\} + \mathrm{I}(x = 0) + \exp\{-\delta \times x\} \times \mathrm{I}(x > 0),$$

where + separates variables and the parameters γ, δ capture the rates of exponential impact decay. The exponential form assumes that impact decay is proportional to the impact level, a common assumption in analyses of water pollution. Compared to linear and linear-quadratic functional forms, this functional form allocates more influence to observations at shorter distances and not vice versa (see Section C1 of the Appendix for details).

3.2.2. Identifying assumption

Our key identifying assumption posits no other discontinuous changes along the river that impact vegetation at the mine-basin. This is supported by the nature of our observations. River basins are defined by natural geographic features that vary smoothly over space, and the granular basin level minimizes systematic differences between adjacent basins.

To validate our approach, we conduct several robustness checks. We control for extensive basin characteristics (geophysical, meteorological, and socioeconomic), use these variables as outcomes to test for spurious discontinuities, and implement covariate matching to ensure that up- and downstream basins are comparable. We also vary the sample considered by excluding mine-basins, restricting to first-order basins, subsetting to narrower periods, and randomizing the treatment assignments as a placebo.

[16] Of the main set of covariates, only the meteorological variables are time-varying; the other variables are time-invariant (geophysical) or fixed at pre-period levels (socioeconomic).

[17] We let $f(x)_{\mathcal{J}} = \mathrm{I}(x \in \mathcal{J})$ for $\mathcal{J} \in \{-10, \dots, -4, 0, \{1, 2, 3\}, 4, \dots 10\}$, and focus on $\beta_{\{1,2,3\}}$.

4. Results

Our analysis reveals that mines significantly reduce vegetation health in downstream areas through water networks. We begin by focusing on effects in the immediate vicinity of mines, i.e., downstream basins up to order three, for which causal identification is strongest. In Section 4.1, we investigate potential heterogeneities in these impacts to better understand the underlying mechanisms. Next, we extrapolate beyond the immediate vicinity and investigate the reach and decay of effects in Section 4.2. Finally, we assess the robustness of our estimates in Section 4.3.

Table 2 presents estimates of the impact of mines on general (left columns) and cropland-specific (right columns) vegetation downstream, measured via a peak vegetation index, based on the EVI. For each outcome, we present estimates from both a plain specification without covariates and a fully saturated specification with controls. The reported estimates represent the causally identified average treatment effect (ATE) for downstream areas, i.e., the difference in vegetation health between downstream (treated) and upstream (control) basins at equivalent distances from mine-basins.

The upper panel of Table 2 shows results for individual basins up to order five, while the lower panel pools the first three downstream basins to provide a more statistically powerful and readily interpretable estimate of the immediate effect. Note that we do not report and interpret estimates for the mine-basin itself, as they are not identified by our research design. Complete results, with estimates for all basin orders and covariates, are available in Table E5 in the Appendix.

The treatment indicators show statistically significant negative effects on vegetation downstream of mines compared to vegetation upstream for both indices of general and cropland-specific vegetation. The (gradual) inclusion of covariates changes estimates only marginally (see Table E5 for extended results), alleviating concerns that findings are driven by confounding factors. Estimates suggest that the effect permeates the immediate vicinity and may extend beyond this area with a similar magnitude, although the precision and statistical significance decline with distance.

For general vegetation, the estimated ATEs for the first three basins downstream of the mine are −0.0057 and −0.0056 (with and without covariates) and highly significant ($p < 0.01$). This is mirrored by estimates for individual basins, where the ATEs for the first three basins are significant ($p < 0.05$) and range from −0.0043 to −0.0087. Estimates for the subsequent basins, including the fourth and fifth downstream basins, are consistently negative but show a drop of statistical significance and a minor drop in magnitude. The reductions in the immediate vicinity correspond to a 1.28–1.35% decrease relative to the sample mean. The impacted area, i.e., vegetation in the first three downstream basins, stretches across 255,000 km^2.

For croplands, the estimated ATEs are slightly larger. For the first three basins downstream of the mine, estimates are −0.0064 and −0.0068 (with and without covariates) and significant ($p < 0.05$). Individual-level estimates for these basins range from −0.0050 to −0.0099 and are significant ($p < 0.05$) with one exception. Results for higher-order basins are similar to those for maximum vegetation, but the drops in magnitude and precision of estimates is larger. Compared to the sample mean, these estimates imply an index reduction of 1.38–1.47% over an affected cropland area of 74,000 km^2.

4.0.0.1. Contextualizing impacts. These effect sizes align with other studies using vegetation indices to approximate agricultural productivity across various contexts, which we discuss in Section 5. To translate these reductions in peak vegetation indices into agricultural terms, we correlated our measure with direct measures of agricultural productivity from LSMS-ISA and AReNA-DHS (see Appendix B2). Based on the AReNA-DHS data, being immediately downstream of a mine is associated with a 0.613% decrease in cereal yields, while the LSMS-ISA data indicates a 1.102% decrease.

Table 2
Main estimation results.

Outcome	*Peak vegetation*		*Peak cropland Veg.*	
(Specification)	(Plain)	(Full)	(Plain)	(Full)
Individual order				
Downstream (1st)	−0.0045***	−0.0043**	−0.0051**	−0.0050**
	(0.0017)	(0.0018)	(0.0025)	(0.0025)
Downstream (2nd)	−0.0049**	−0.0048**	−0.0058*	−0.0067**
	(0.0022)	(0.0024)	(0.0031)	(0.0032)
Downstream (3rd)	−0.0085***	−0.0087***	−0.0088**	−0.0099***
	(0.0028)	(0.0029)	(0.0037)	(0.0038)
Downstream (4th)	−0.0049*	−0.0062*	−0.0029	−0.0044
	(0.0030)	(0.0033)	(0.0038)	(0.0040)
Downstream (5th)	−0.0034	−0.0053	0.0007	−0.0016
	(0.0033)	(0.0037)	(0.0042)	(0.0045)
Pooled order				
Downstream (1st–3rd)	−0.0057***	−0.0056***	−0.0064**	−0.0068***
	(0.0018)	(0.0020)	(0.0025)	(0.0026)
Fit statistics				
Observations	110,576	110,524	93,036	93,000
# upstream basins	6,084	6,084	5,232	5,232
# downstream basins	5,899	5,896	4,769	4,767
R^2	0.903	0.908	0.816	0.822
Controls				
Geophysical	No	Yes	No	Yes
Meteorological	No	Yes	No	Yes
Socioeconomic	No	Yes	No	Yes
Fixed-effects				
Year (2016–2023)	Yes	Yes	Yes	Yes
Mine	Yes	Yes	Yes	Yes

Note: The table reports estimates of the average treatment effects by equidistant basin order based on Eq. (1). The left columns hold results for the overall peak vegetation index; the right columns for the cropland-specific peak vegetation index. The first and third columns include no covariates, whereas columns two and four include the full set of control variables. Estimates in the upper panel correspond to the average effect at individual orders, while the lower panel reports a pooled estimate for the three basins that are immediately adjacent to the mine-basin. All specifications include mine and year fixed effects. Standard errors are (in brackets) and clustered at the mine level; significance levels are indicated as ***: 0.01, **: 0.05, *: 0.1.

The scale of these impacts becomes apparent when considering the affected area. The first three downstream basins cover 280,000 km^2, of which 74,000 are croplands — an area equivalent to Ghana's total cropland, falling between the cropland areas of the United Kingdom and Malaysia. Using average yields from both datasets and scaling up for all affected croplands, the estimated yield reductions translate into annual production losses of 91,000–205,000 tons of cereals. To contextualize these losses, the World Food Programme distributed approximately 1.7 million tons of food aid to over 38 million recipients across African countries in 2023. Our estimates of agricultural production losses from water pollution downstream of mines — covering just the immediate vicinity of mines — correspond to 5.4 to 12.1% of this major humanitarian food aid program.

4.1. Heterogeneity

Next, we investigate how our results vary along several dimensions to identify where mining impacts on vegetation and agriculture are most pronounced. We examine heterogeneity related to the characteristics of basins (biome, regions, and suitability for crop cultivation) as well as mine characteristics (total mining area, activity, and commodity type).

Fig. 4 provides an overview of the results. We present specifications with full controls and pooled estimates for the first three basins, allowing for full heterogeneity by re-estimating with different subsets of the sample. Estimation results are available in Tables E7 and E8 of the Appendix.

4.1.1. Environmental heterogeneity

We first analyze spatial heterogeneities of basins, which are visualized in the upper half of Fig. 4. To check for differences across biomes, we assign each basin to one of three broad ecological groups — grasslands, forests, and deserts — that are aggregated from the more granular ecoregions of Dinerstein et al. (2017). We find negative effects for both the grassland and forest biomes. Grasslands constitute 69% of the vegetation sample and 74% of the cropland sample, while forests make up 15% and 17%. The effect sizes are comparable to the baseline, although impacts in forest biomes appear somewhat larger. In desert biomes (representing 15% and 8% of the sample), we find no significant effect on either the cropland or overall peak vegetation, likely reflecting limited vegetation to be impacted in these regions.

For regional differences, we separate samples based on the classification system of the USDA.[18] We find substantial effects in West Africa, which represents 29% of the vegetation sample and 26% of the cropland sample. In Southern Africa (61% and 64%) or North & East Africa (10%), we find no significant effects. This regional disparity likely relates to mining practices and commodity types — artisanal gold mining is particularly common in West Africa. Additionally, our identification strategy may reach its limits in the case of South Africa, where a vast cluster of mines permeates entire basin systems, complicating the isolation of downstream effects.

The agricultural potential of affected regions captures another important dimension of impacts. We follow the GAEZ framework in categorizing basins into high, medium and low crop suitability groups, based on the maximum suitability across 30 crops. We find significant negative effects on basins with high crop suitability, which comprise 36% of the vegetation and 39% of the cropland sample. The point estimates for this sample exceed the baseline for both the general and cropland-specific vegetation. In contrast, estimates for medium (43% and 47%) and low suitability (22% and 13%) are smaller and statistically insignificant. This pattern is concerning because more than 35,000 km^2 (over a third) of highly suitable land in the immediately affected basins is actively used for crop production, with implications for subsistence farming and food security.

4.1.2. Mine heterogeneity

Turning to mine characteristics, we first examine whether mine size influences downstream impacts. We progressively restrict the sample to mine-basins with mining areas of at least 0.5 km^2, 1 km^2, and 2.5 km^2, which reduces the sample to around 51%, 40%, and 25% of its original size. For overall vegetation, we observe an increase in effect size as mining area increases. For the 2.5 km^2 cutoff, the estimate is 50% higher compared to the baseline. This result aligns with expectations, as larger mines typically produce more discharge material and cause stronger contamination of water systems. For croplands, however, we find imprecise estimates with no discernible trend in magnitudes.

For mining activity, we use growth as a proxy and analyze subsets of active mines that exhibited (i) any growth, (ii) at least 10% growth, or (iii) at least 25% growth over the observed period. Our results show no substantial heterogeneity along this dimension for either natural vegetation or the cropland vegetation. Effect estimates remain stable, although precision decreases with smaller sample sizes (approximately 67%, 63%, and 55% of the baseline). This suggests that our dataset

[18] The system seeks to reflect differences in crop types and cycles. The regions are West Africa (with the Burkina Faso, Guinea, Mali, Ghana, Nigeria, and Niger as the most represented countries with over 250 basins), Southern Africa (with South Africa, Zimbabwe, Namibia, Tanzania, Botswana, Angola, the DRC, Zambia, and Mozambique), and North & East Africa (where only Morocco exceeds 250 basins), which we pool.

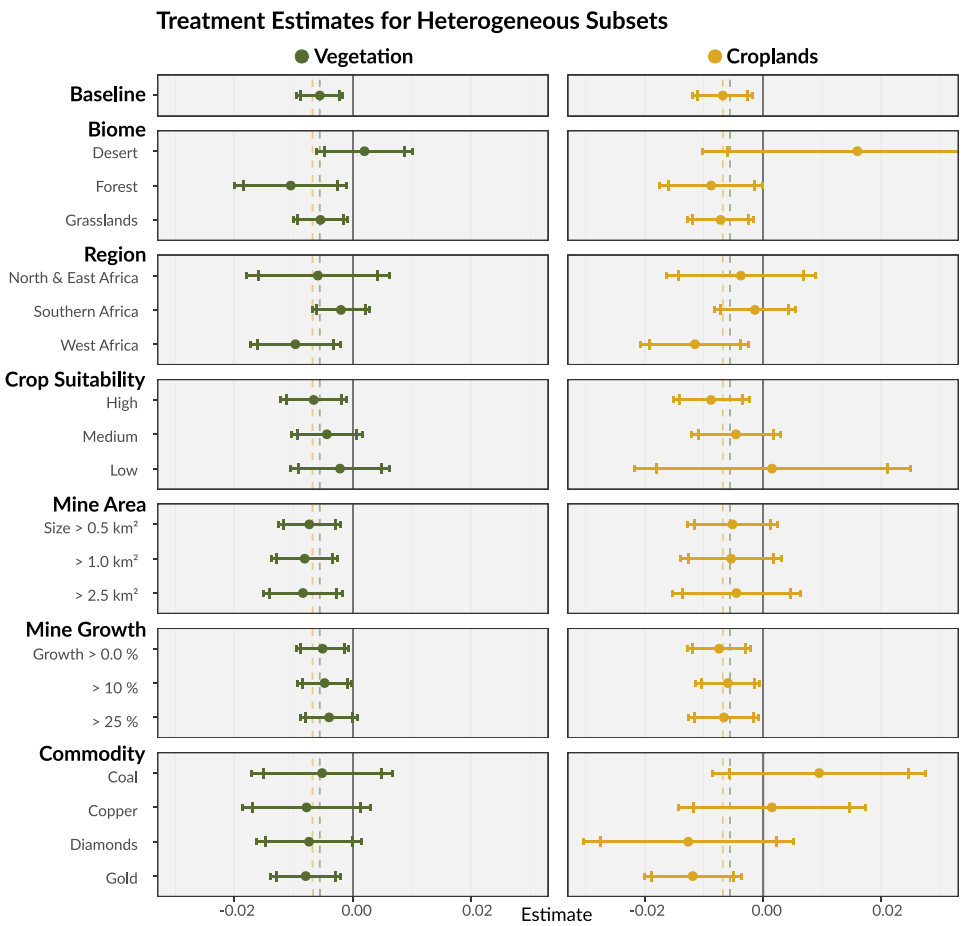

Fig. 4. Dots indicate the average treatment effect in the first three basins; whiskered lines indicate 90% and 95% confidence intervals. Dashed vertical lines indicate point estimates for baseline specifications.

of mine locations, which is anchored in 2019, adequately captures the relevant impacts of mining sites regardless of their status. The lack of heterogeneity may also reflect offsetting factors: expanding mines might implement better precautions, while inactive, poorly maintained sites could still produce considerable pollution.

Finally, we examine how impacts vary by commodity types. We focus on gold, diamonds, copper, and coal — commodities that are well-represented in the data and which can reasonably be isolated from other commodities. Results reveal a strong negative effect for gold mining, which we isolated in 26% of the vegetation and 29% of the cropland sample. Compared to the baseline, point estimates for gold mining are 40% larger for vegetation overall and 75% larger for croplands. Among the other commodities — diamonds (13% of the sample), copper (6%), and coal (8%) — only diamond mining shows marginally significant negative effects on overall vegetation.

4.2. Impact decay

Having established significant and economically meaningful downstream impacts of mining in the immediate vicinity, an important question remains: How far downstream do these effects persist? The spatial extent of effects is crucial for understanding the overall impact, gauging the size of the externality, and designing effective mitigation strategies. While critically important, this analysis extends beyond our identification strategy — at greater distances, basins may diverge in characteristics and results cannot be interpreted as strictly causal. Our analysis here provides suggestive evidence and highlights the potential overall stream-mediated impact of mining.

The range of impacts is related to the types of mediators that it arises from. Vegetation is affected by various types of water pollution from mining, including heavy metals, acidification, and salinity (see Appendix B1 for further details). While contaminants gradually disperse along streams, buildup and tipping points could lead to more persistent impacts. Acidification, for instance, is initially triggered by pollution from mines but is sustained by extremophile microbes. We empirically investigate the decay of impacts by (1) extrapolating our analysis beyond the immediate vicinity, and (2) imposing an exponential decay functional form to estimate the speed of decay.

Fig. 5 visualizes estimates along basin orders, comparing all basins to the one immediately upstream of the mine (order -1).[19] If mines had no impacts, we would expect flat estimates across all basin orders. Non-directed mine impacts (e.g., from air pollution) would appear as a V-shape, where the right 'wing' of the V would be shifted by downstream effects and rotated if these effects decay with distance.

The figure reveals three notable patterns. First, negative impacts persist beyond the immediate vicinity of mines; effects appear to increase in size after the sixth basin. Second, trajectories differ slightly between general vegetation and croplands. Croplands suffer slightly larger impacts near mines but show more decay at larger distances. Impacts on general vegetation are more persistent, yielding significant negative estimates even at large distances. Third, there is an indication

[19] The visualization differs from the coefficients in Table 2 and Table E5 (which report differences between equidistant up- and downstream basins, and not deviations from the first upstream basin) to better convey changes from basin to basin.

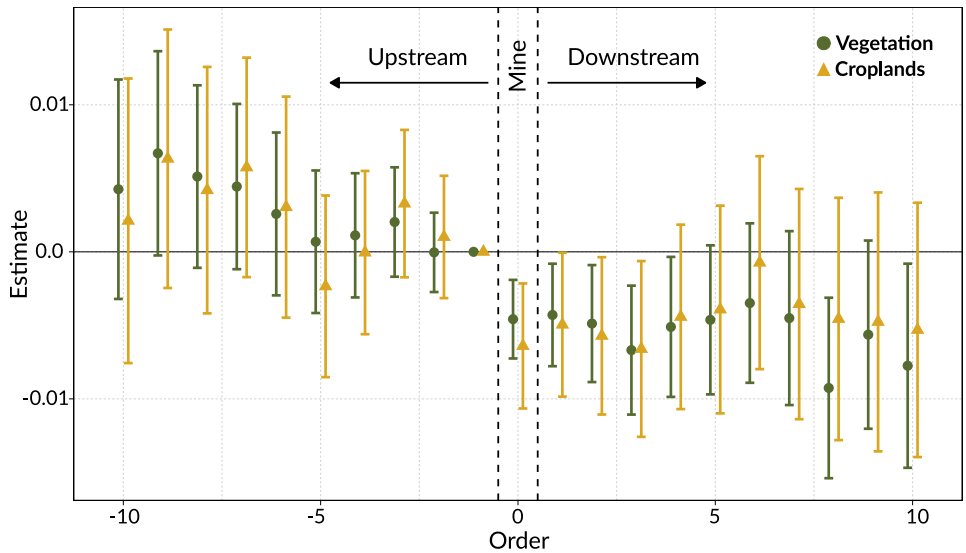

Fig. 5. Estimated order coefficients for all up- and downstream basins (with the mine-basin in the center) for the overall peak vegetation index (green circle) and the cropland-specific index (yellow triangles) with full covariates. Whiskers represent 95%-confidence intervals. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of a V-shaped pattern upstream, with the right side being rotated and shifted downward.

To further investigate decay patterns, we apply an exponential distance decay function to the downstream distance, which aligns with hydrological models of contaminant dissipation. The detailed results appear in Appendix C1.1. In addition to the strength of the effect, we estimate the rate of decay and allow it to vary from meter to kilometer scales. We find minimal decay over our sample range, and overall impacts that are comparable to the immediate vicinity of mines. The effect on overall vegetation downstream begins at −0.0062 and halves after 281 km, which extends beyond the support of our sample. For croplands, the effect begins at −0.0068 and decays by half at a distance of 72 km, diminishing to 10% of its initial value at 234 km.

These findings highlight that vast areas downstream of mines may be impacted by their water pollution, extending considerably beyond the immediate vicinity examined in our main analysis. Isolating the nature of these effects and identifying specific mechanisms driving them is an important avenue for future research.

4.3. Robustness

To ensure the validity of our main findings, we conduct a comprehensive set of robustness checks that address potential concerns about our empirical strategy. We organize these robustness checks into six categories: (i) included covariates, (ii) outcome measures, (iii) sample definition, (iv) unobserved factors, (v) covariate balance, and (vi) placebo tests. Fig. 6 visualizes estimates under various checks; the complete results appear in the Appendix.[20]

Covariates. First, we examine whether our results are sensitive to the choice of control variables. We test four variations: (1) adding controls for air pollution based on PM2.5 concentrations, (2) controlling for conflict intensity, (3) using distance to coast as a geographic control, and (4) using alternative meteorological data for rainfall and temperature. None of these modifications substantially alter our baseline findings, suggesting that our results are not driven by omitted variables related to these factors.

Outcome measures. Next, we assess whether our results depend on our specific choice of outcome variables. We (1) substitute the EVI with the Normalized Difference Vegetation Index (NDVI), and (2) use alternative land use masks for vegetation and cropland-specific vegetation index.[21] We also (3) construct the peak EVI to reflect pixel-level peaks (accounting for varying seasonality) before aggregation to the basin level, and (4) use the mean annual EVI to reflect vegetation over the full year, instead of the peak value. These variations change the magnitude of estimates due to different sample moments, but the qualitative results remain consistent, showing significant negative effects of mines on downstream vegetation.

Sample definition. We also verify the robustness of our results across different sample restrictions. First, we focus on periods around the 2019 satellite imagery used in mine delineation: (1) 2019 and after, and (2) a narrower 2018–2020 window. Second, we modify the sample by (3) only including basin systems with at least one up- and downstream basin, (4) excluding mine-basins entirely, and (5) analyzing only immediate first-order basins of mine-basins that have at least one up- and downstream basin to ensure we recover the intended variation. Finally, we also consider a specification where we average outcomes and covariates over all years (see Column 2 of Table E11). While the precision of estimates decreases with smaller samples, our key finding remains robust.

Unobserved factors. We further account for potentially confounding, unobservable factors through alternative fixed effects structures. We introduce hierarchical fixed effects at the (1) Level 6 and (2) Level 8 super-basins, which contain an average of 67 and 6 of our Level 12 basins. We also (3) add country-by-year fixed effects to control for changing national conditions, and (4) include mine-specific linear time trends. In addition, we vary the clustering of standard errors to the mine and year level, the mine-by-year level, and the individual basin level (see Columns 6–8 of Table E12). These specifications yield consistent results, indicating that our findings are not driven by unobserved spatial or temporal factors.

[20] Further details are provided in Appendix C3; Tables E9 to E12 report the full estimates.

[21] We use a narrower version for the vegetation mask that excludes sparse and flooded vegetation, and an Africa-specific cropland mask that is anchored in 2019 (Digital Earth Africa, 2022).

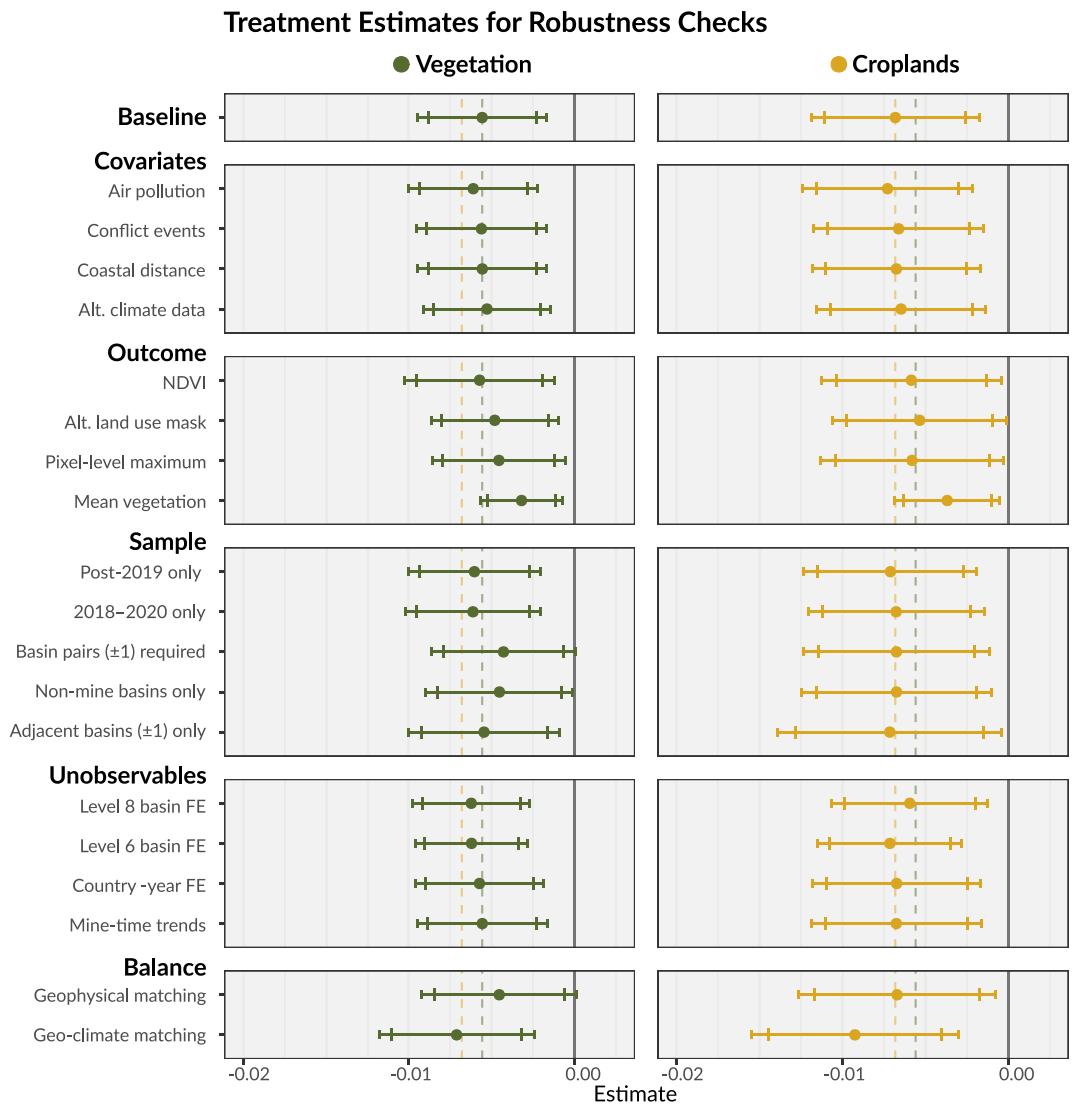

Fig. 6. Dots indicate the average treatment effect in the first three basins; whiskered lines indicate 90% and 95% confidence intervals. Dashed vertical lines indicate point estimates for baseline specifications.

Covariate balance. To address potential imbalance between treatment and control groups, we use coarsened exact matching (Iacus et al., 2012). We match upstream and downstream basins on (1) elevation, slope, and soil type in a basic specification, then add (2) temperature, and precipitation in an extended version. The matched samples achieve excellent balance on observables (see Figure D12 and Figure D13 in the Appendix), and the resulting estimates are significant and of similar magnitude. This further corroborates the validity of our identification strategy.

Placebos. Finally, we conduct two additional validation exercises. First, we randomly reassign the treatment status across basins and re-estimate a total of 5000 times, finding no evidence of the impacts observed in our actual analysis (see Appendix C3.2 for details, and Figure D11 for results). Second, we use our control variables as well as nighttime light emissions and forest loss as placebo outcomes. We find no structural differences between up- and downstream basins (see Appendix C3.3 for details), confirming that our identification strategy isolates mining impacts rather than pre-existing environmental or social differences.

5. Discussion

Our results provide strong causal evidence for water-mediated effects of mines on vegetation and agriculture. We find reductions of peak vegetation indices of 1.28–1.35% for all vegetation and 1.38–1.47% for croplands specifically. Below, we discuss the mechanisms behind these impacts, how to interpret and gauge their size, which limitations to consider, and what our results imply for policy and future research.

5.1. Mechanisms

Our research design identifies mining impacts on vegetation that are specifically mediated along water streams, separating them from other effects of mining. The main mechanisms explaining our findings are water pollution and its second-order effects, such as human responses to it. Water pollution emerges as the primary mechanism behind our observed effects. As a major and well-documented externality of mining, it provides the most direct explanation for reduced vegetation health downstream of mines. While our design provides only indirect evidence for specific mechanisms, we analyzed available water quality data (from United Nations Environment Programme, 2025) to confirm the role of water pollution (see Appendix B3).

The data reveal markedly lower water quality in mine-basins and downstream areas compared to upstream basins. Key indicators are consistent with mine-induced pollution: electrical conductivity (measuring dissolved salts and minerals), sodium discharge and adsorption, and sulfate concentrations that can cause acidification are increased

downstream. However, these water quality data have important limitations: samples are exclusively from South Africa, were obtained at irregular intervals using different methods, and measurements of heavy metals are lacking. More comprehensive data are needed to assess water quality for the entire continent and differentiate between pollutants, for instance, from mercury- or cyanide-based gold extraction (see Verbrugge et al., 2021).

Our results could also be influenced by secondary mechanisms in response to water pollution, such as soil pollution, water scarcity, and human impacts. Water systems and soils are closely interlinked (Cheng et al., 2021), and riverine pollution affects both groundwater and soils, impacting the rainfed agriculture that is common in Africa (see Appendix B1.1 for more details). Similarly, mines and water pollution (as well as increased evapotranspiration and percolation) creates water scarcity that may exacerbate the overall downstream impacts. By contrast, the net effects of human responses are ambiguous, as locals may be affected and respond in ways that could amplify or attenuate the direct impacts of mines.

Evidence for air pollution suggests complex human impact patterns. While plant growth is primarily affected directly, pollution also impacts farmer health and consequently labor supply, productivity, income, and land allocation (see Fugiel et al., 2017; Kotsadam and Tolonen, 2016; Miao et al., 2017). To adapt to water pollution, farmers may shift towards more resilient crops (that tolerate, e.g., higher salinity levels) and attenuate overall impacts or respond by farming less intensively or out-migrating, amplifying impacts (note that Aragón and Rud, 2013, 2015, do not find evidence for crop diversification or migration in more focused studies).[22] When water sources are contaminated, agricultural practices could also be modified by reducing irrigation — which is uncommon in the region — or deciding against adopting irrigation systems in the first place.

5.2. Interpretation

Our analysis reveals mines' considerable downstream impacts on vegetation and agriculture. These effects are mediated by water pollution and adaptation responses, making them likely to accumulate and persist for many years. We find that downstream of mines, the peak cropland vegetation index is reduced by 1.38–1.47% of the sample mean.

These effect sizes align with several other studies that use vegetation indices to approximate agricultural productivity. To provide context for our findings, we briefly summarize related results based on similar satellite-derived measures. Using peak annual EVI, Wuepper et al. (2023) find a 2.2% change per step on a ten-step institutional quality index over five years. In India, Asher and Novosad (2020) estimate a 3.5% increase from new road construction, while Ghosh and Vats (2022) find that a 10% increase in guaranteed income beneficiaries increases the same measure by 8.1%. Information and communication technologies show similar effects: Gupta et al. (2024) find that districts in India with one-standard-deviation higher shares of non-state language speakers generate 0.6% lower yields. Agarwal et al. (2024) find that access to 4G internet increased EVI-implied yields by 12.5% after six years.

Studies examining water and agricultural practices provide more direct comparisons to our findings. Asher et al. (2022) estimate that villages with access to irrigation canals show 1.7% and 7.1% higher EVI-proxied yields in the monsoon and winter seasons, respectively. Boudot-Reddy and Butler (2025) find that one-standard-deviation increases in groundwater extraction increase yields by 8.6% in winter. Deines et al. (2019) find that long-term conservation tillage increases satellite-derived yields by 0.7 to 3.3% by preventing soil degradation. By comparison, Jain et al. (2019) find a 4.5% yield gain from more effective fertilizer application.

We translated our results into agricultural terms by correlating our measure with LSMS-ISA and AReNA-DHS data. For the immediately affected 74,000 km^2 of smallholder croplands that are up to around 33 km downstream of mine-basins, we estimated annual losses of 91,000–205,000 tons of cereals. These losses correspond to 5.4 to 12.1% of the food aid distributed annually by the World Food Programme.

When interpreting these results, two important caveats apply. First, they reflect only one specific pathway among the many externalities of mines. Second, they might be partly offset by redistributing the benefits from mining. Mining also affects agriculture, vegetation, and livelihoods via air pollution and other non-directed factors that our research design does not capture. This includes positive effects, such as increased local incomes and wealth ownership (Aragón and Rud, 2015; von der Goltz and Barnwal, 2019) and alternative employment opportunities (Kotsadam and Tolonen, 2016; Gräser, 2024). Meanwhile, water pollution from mines creates numerous additional negative effects beyond agriculture, including water scarcity, biodiversity loss, and detrimental impacts on human health and cognition (Gittard and Hu, 2024; Van Vliet et al., 2017; Vörösmarty et al., 2010).

5.3. Limitations

Our analysis faces three key limitations that warrant discussion. These are related to uncertainty in the outcome measures, incomplete mining data, and limited mechanistic evidence.

Outcome measures. We rely primarily on remotely sensed vegetation indices, which introduces measurement noise that affects our estimates. While these indices offer consistent and highly granular measures of vegetation health that enable our analysis, they have practical and inherent constraints. Our peak vegetation indices perform well as a general proxy, but cannot provide a complete picture of complex vegetation patterns. No single variable can capture the heterogeneity of impacts across different crop types, soil conditions, and climatic zones (Bolton and Friedl, 2013), and data for calibration is limited. When we relate our outcome measure to local agricultural survey data, this introduces additional noise and likely produces attenuated estimates. Future research with a narrower geographical focus could address this limitation by combining remote sensing with more detailed ground-truth agricultural data.

Incomplete mining data. While we use a comprehensive dataset of mine sites with verified locations and surroundings, these data have important gaps. Critical information about mining sites and operations — including their status, activity, production methods, and commodities mined — is not directly available. We partially address this limitation by assessing mining area growth over time and adding commodity information through supplementary sources. However, the data does not distinguish between artisanal and industrial mining, which differ substantially in management practices, processing methods, and environmental impacts. While geological bedrock characteristics can be used to proxy for artisanal gold mining presence (Girard et al., 2022, 2025), the frequent co-location of artisanal and industrial mines means that our research design cannot cleanly disentangle effects. Conducting more targeted analyses that integrate detailed information about mining operations represents an important avenue for future research.

[22] While our design only reflects the net effect, we expect the role of adaptation to be constrained and attenuate overall impacts. A rational response to environmental change would be to mitigate it where possible, but other scenarios are thinkable. If, for instance, impacts drive downstream farmers to abandon their croplands and migrate, the overall effect will be amplified by the decrease in labor supply. If their destination were upstream — which is unlikely, given the other undirected impacts of mines — impacts would be exacerbated further.

Mechanistic evidence. Our study provides suggestive rather than definitive evidence for the exact mechanisms behind the water-mediated effects. Water quality and pollution data are scarce, geographically patchy, and methodologically inconsistent across regions (Jones et al., 2024). We investigate heterogeneity across commodity types, regions, and mine growth patterns to infer potential mechanisms but cannot provide direct evidence linking specific mining activities to vegetation impacts. Additionally, while our research design identifies local treatment effects, it cannot establish the full geographic reach of these impacts. More in-depth analyses of mechanisms and the reach of effects are left to future work.

5.4. Outlook

Our results have important implications for future research and policy. Two key priorities emerge: overcoming the data scarcity that constrains both scientific understanding and policy responses, and implementing targeted interventions to mitigate mining's external costs and distribute its benefits.

A common theme across our analysis is the scarcity and relatively poor quality of available data. Addressing this gap is paramount, as improved information can advance research and enable policymakers to design effective mitigation strategies. Agricultural statistics remain challenging to collect for smallholder and subsistence farming, limiting our understanding of farm-level impacts. Information on mining sites is inadequate and poorly accessible (Maus and Werner, 2024), especially for artisanal mines where data dissemination must safeguard against potential human rights abuses. Water pollution data are similarly essential but scarce (Jones et al., 2024).

Remote sensing technologies offer considerable potential to address these data gaps. For water pollution, satellite imagery can detect large spills and tailings dam failures (Rudorff et al., 2018; Ruppen et al., 2023), while refined approaches may help identify heavy metal contamination or sediment concentrations (Swain and Sahoo, 2017). Automated monitoring of mine sites using machine learning methods (Sepin et al., 2025) could reduce laborious mapping tasks. However, these remote sensing approaches depend on high-quality ground-based data for calibration. Where institutional capacities for producing such data are lacking, community-based initiatives, for instance, for water monitoring, could serve as an inexpensive and effective complement (Ruppen et al., 2021).

Policymakers should address mining's external costs through targeted interventions. Mining concessions should explicitly recognize externalities and factor in impacts on local economies, vegetation, and food security. Our results provide concrete evidence for including water-mediated impacts in concession allocation decisions. Formalizing artisanal mining through official titling could increase miners' incentives to invest in precautionary equipment, reducing impacts at the source with benefits for human health and the environment.

Supranational interventions are essential for effectively addressing mining externalities by shifting regulatory burdens from mineral-producing countries towards all who benefit from the extracted resources. The Global Industry Standard on Tailings Management (GISTM) presents an important first step for international governance of industrial mines. Our findings highlight that persistent water pollution, and not just catastrophic failures of tailings dams, requires attention. The Minamata Convention targets the use of mercury, which is particularly damaging and still used in processing gold — the commodity for which we find the strongest downstream impact. Supply chain measures could also help reduce minerals sourced through harmful practices, following models like the European Union's Deforestation Regulation (EUDR) that applies to companies selling products on the EU market.

6. Conclusion

In this paper, we identified the causal effects of mining on agricultural productivity mediated by water. Using a quasi-experimental research design, we exploited the discontinuity created by mine sites along directed river networks for identification. We compared differences in agricultural productivity — measured through remotely sensed peak vegetation indices — upstream and downstream of mine locations. Our main specification revealed reductions in peak vegetation of 1.28–1.35% for general vegetation and 1.38–1.47% for croplands up to approximately 33 km downstream of mines. This corresponds to annual losses of 91,000 to 205,000 tons of cereals across 74,000 km^2 of immediately affected croplands.

Our estimates capture only one component of the total external costs of mining, and for agriculture specifically. They do not reflect impacts that are not mediated along river networks, such as effects on air pollution or income. Additionally, our indirect measurement of agricultural productivity may attenuate estimates of the water-mediated effects. Despite these limitations, our findings contribute important evidence to discussions about resource extraction in Africa, particularly in regions with weak environmental governance.

The documented effects highlight the urgent need for interventions that reduce mines' negative impacts on water systems and distribute their costs and benefits. Industrial and informal mining operations should be required to implement proper containment facilities and reduce or substitute particularly harmful methods. Enhanced monitoring of mines and water quality is necessary to address the data limitations we encountered to provide a basis for understanding impacts and guiding effective policies.

This study opens several promising avenues for future research. While a comprehensive Africa-wide mine dataset enables broad analysis, it lacks detailed information on individual mine characteristics. Future studies that incorporate data on containment facilities, mine types, and specific pollutants would allow for more precise impact analyses. Similarly, spatial data on crop distributions could enable detailed investigations of which crop types are most susceptible to mining-induced pollution and inform adaptation strategies. Alternative research approaches might address questions our design could not fully answer, such as disentangling the impacts of industrial versus artisanal mines or providing stronger evidence on effect decay. Such research would further strengthen the evidence base for targeted interventions that help balance the economic benefits of mining with its environmental and human costs.

CRediT authorship contribution statement

Lukas Vashold: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization, Supervision . **Gustav Pirich:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization . **Maximilian Heinze:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization . **Nikolas Kuschnig:** Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization, Supervision.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jdeveco.2025.103671.

Data availability

Code and data are provided in the replication package at https://github.com/maxmheinze/minesriversyields.

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